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## Tradeoffs between nutrient load reductions and greenhouse gas dynamics in constructed agricultural wetlands

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E-mail: [martyn.futter@slu.se](mailto:martyn.futter@slu.se)**Keywords:** nature based solutions, ponds, free surface wetlands, tradeoffs, methane, nutrientsSupplementary material for this article is available [online](#)

## Abstract

Perceived tradeoffs between ecosystem services (ES) delivered by constructed ponds and wetlands in agricultural landscapes may limit their widespread uptake for environmental management. These nature-based solutions (NBS) can mitigate downstream eutrophying effects of agricultural nutrient runoff and contribute to carbon (C) storage. However, they can also be significant greenhouse gas (GHG) sources. Here, we report on water chemistry, dissolved GHG concentrations, total methane (CH<sub>4</sub>) emissions, and sediment properties measured over three years at 40 Swedish free-surface constructed agricultural wetlands (ponds). Large temporal variations in inlet water chemistry highlighted the influence of seasonality and land management. Inlet phosphorus (P) concentrations were positively correlated with water column dissolved CH<sub>4</sub> and sediment P concentrations; a clear tradeoff in nutrient retention *vs.* climate impact. A more nuanced pattern occurred for nitrogen (N) where inlet concentrations correlated positively with dissolved nitrous oxide (N<sub>2</sub>O) concentrations. However, these wetlands mitigated downstream dissolved N levels as suggested by lower outflow concentrations. All waterbodies were supersaturated with CO<sub>2</sub>. Higher CO<sub>2</sub> and N<sub>2</sub>O concentrations occurred during fall/winter than summer. The opposite pattern occurred for CH<sub>4</sub> fluxes and dissolved CH<sub>4</sub> although relative contributions from ebullition were greater during fall/winter. Sediment C concentrations were unrelated to any measured parameters, suggesting it would be difficult to purposefully design wetlands for C sequestration. Although there are tradeoffs between mitigating downstream eutrophication and climate impacts, this should not preclude the use of constructed free surface wetlands as multifunctional NBS for ES delivery in agricultural landscapes.

## 1. Introduction

Climate change and eutrophication are amongst today's most pressing environmental problems (Rockström *et al* 2024). Measures to mitigate climate warming and to limit excessive transport of nutrients from land to receiving waters are urgently needed. Nature based solutions (NBS) work with nature and

natural processes to deliver a range of ecosystem services (ES; Ferreira *et al* 2023) including mitigating climate impacts (Seddon *et al* 2020), eutrophication control (Djodjic *et al* 2022, Rizzo *et al* 2023) as well as cultural and biodiversity supporting services (Hambäck *et al* 2023).

Constructed free surface wetlands and ponds (referred to hereafter as 'wetlands') are one type of

NBS used for mitigating eutrophying effects of agricultural runoff (Djordjic *et al* 2022, Nilsson *et al* 2023, Rizzo *et al* 2023, Hermansen *et al* 2026). The goal of mitigation is to prevent nutrients (primarily nitrogen (N) and phosphorus (P)) in agricultural runoff from reaching downstream water bodies where they can cause excessive and detrimental aquatic plant growth, leading to eutrophication. Mitigation can occur through two processes: retention and removal. Retention stores nutrients temporarily (e.g. in vegetation or sediments) while removal permanently transports nutrients out of the wetland through, e.g. evasion to the atmosphere, sediment dredging, or vegetation harvesting. The latter two interventions are part of routine wetland maintenance. Carbon (C) retention through sequestration in wetland sediments has positive climate impacts. However, C removal through evasion to the atmosphere as carbon dioxide (CO<sub>2</sub>) or methane (CH<sub>4</sub>) has negative climate impacts. Over time, retained C and nutrients may be re-mobilized through, e.g. scouring during high flow events, or chemical release from sediments under anoxic conditions. While potential eutrophication control benefits of wetlands are well established, their role in C cycling is less clear. Although constructed agricultural wetlands can retain C in their sediments (Ljung and Lin 2023), they can also be disproportionately large emitters of greenhouse gases (GHGs) (Malerba *et al* 2022) including nitrous oxide (N<sub>2</sub>O), CH<sub>4</sub> and CO<sub>2</sub>.

There is widespread public sector financial support for NBS to deliver ES in, e.g. China (Wang *et al* 2019), Europe (Pe'er *et al* 2019) and elsewhere. State financing of wetlands for ES delivery in agricultural areas has been supported in Sweden since the 1990s (Graversgaard *et al* 2021, Geranmayeh *et al* 2023), and is available for both wetland construction and maintenance. Originally, funds were available for wetlands to reduce N loads from land to sea. Coverage was later extended to biodiversity support, P load reduction, and most recently drought mitigation (Graversgaard *et al* 2021, Geranmayeh *et al* 2025).

The range of potential benefits delivered by constructed agricultural wetlands either individually or across a region is greater than the suite of benefits which can receive financial support. As well as the ES for which they were financed, individual wetlands and wetlandscapes comprised of multiple wetlands in a region can deliver other benefits including biodiversity support, recreational and cultural ES. Both Hambäck *et al* (2023) and Whitfield *et al* (2024) highlight the need to consider multifunctionality at both the individual wetland and landscape scales to characterize the range of ES and disservices delivered.

Despite strong state support for wetland creation, researchers have expressed concerns about their effectiveness for nutrient load reduction in agricultural areas (Land *et al* 2016, Geranmayeh *et al* 2023). Wetlands do not always deliver the expected

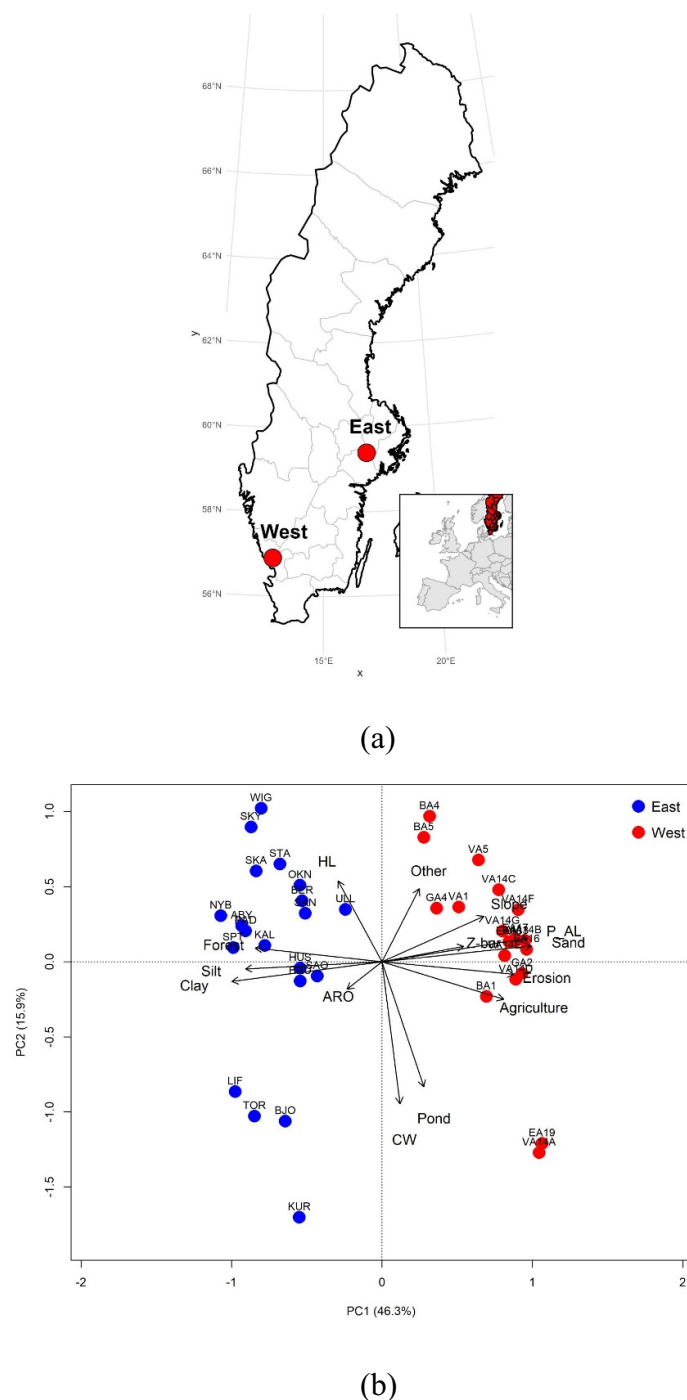
benefits, and may also introduce a form of 'pollution swapping' where eutrophication mitigation is associated with increased GHG emissions (Malerba *et al* 2022); these emissions, which are anthropogenic in nature, can be large and should be included in national GHG inventories (Malerba *et al* 2024, Peacock *et al* 2025).

Here, we assess tradeoffs between nutrient load reduction and climate impacts through an analysis of correlations between water chemistry, GHG dynamics and sediment properties in an ensemble of Swedish constructed agricultural wetlands. We measured inflow and outflow nutrient concentrations, dissolved GHG concentrations, CH<sub>4</sub> emissions, and sediment composition to answer the following questions: (i) are there seasonal differences in the relationship between water chemistry, dissolved GHG concentrations and sediment properties, (ii) are there differences in inlet and outlet water chemistry which could be indicative of in-wetland nutrient processing, (iii) is nutrient load reduction potential of individual wetlands related to GHG dynamics, and (iv) is wetland C storage related to water chemistry or GHG dynamics?

## 2. Study sites

Forty Swedish agricultural wetlands were sampled for water chemistry, GHGs and sediment properties. Contact was established with all current landowners to obtain sampling permission and information about wetland age and maintenance history. Given changes in land ownership, it was not always possible to determine age, but available data indicated wetlands were constructed between 1 and 34 years prior to first sampling. Wetland areas and catchment areas were established using QGIS. Wetland bathymetry was measured using an Ecolode<sup>tm</sup> depth sounder (Lindau 2022).

Wetlands are located in Koppen–Geigen climate zones Dfb (east, Mälardalen, cold, no dry season, warm summer) and Cfb (west, Halland, temperate, no dry season, warm summer) (figure 1(a)). There were 21 wetlands in the eastern region and 19 in the west. Land cover in both regions is dominated by agriculture with forests on less productive soils and a low percentage of urbanization. Western catchments typically have higher percentages of arable land than eastern catchments (respective means of 78% and 59%) while the reverse is true for forest cover (means of 11% and 39%) (figure 1(b), supplementary table 1). There was 11% pasture cover in both regions. Western catchments are steeper with coarser, less erodible soils (figure 1(b), supplementary table 1). On average, western soils are 65% sand and 25% silt while eastern soils are approximately 40% clay and 40% silt. Plant available P (P<sub>AL</sub>, or ammonium lactate



**Figure 1.** (a): Map showing Sweden, the location of the two regions where wetlands were sampled and the location of Sweden in Europe. (b): PCA plot showing catchment and wetland characteristics for the study sites. Red dots represent western (Holland) sites and blue dots represent eastern (Mälardalen) sites. The plot uses symmetric scaling. Sand, silt, clay represent the percentages of different soil particle size classes in the catchments. Land cover percentages are divided between agriculture, forest and other. ARO represents catchment area while CW is the constructed wetland area. HL is the hydraulic load, P<sub>AL</sub> is the P<sub>AL</sub> class and erosion is the erosion class.

extractable soil P) was higher in western than eastern catchments (figure 1(b)). There were no regional patterns in wetland area or hydraulic load (HL and ARO) (figure 1(b)). However, hydraulic load was negatively correlated with percent pasture (figure 1(b), supplementary table 1). Average annual temperature in eastern catchments is 6.5° C versus 8.7° C in the west. Annual precipitation is 576 mm yr<sup>-1</sup> and

856 mm yr<sup>-1</sup> in eastern and western regions respectively (SMHI). Excess nutrient leakage from agricultural runoff is an environmental problem in both regions. Further information about catchment properties is available in Ferguson (2019), Lindau (2022), and Kero (2022).

Water chemistry measurements were made at the main inlet and outlet of the wetland. The former

was identified at the first sampling visit. Parameters included chlorophyll-a, filtered and unfiltered total P, phosphate, ammonium, nitrate, total N, total organic carbon (TOC), suspended matter, dissolved oxygen (DO) concentration and percent saturation, electrical conductivity (EC) and alkalinity. Water temperature and atmospheric pressure were measured at each sampling event.

Water samples for chemistry and dissolved GHG analysis were collected during 2021–2023 and classified into summer (June—first week in September) and fall/winter (late September–February) seasons. Each wetland was visited between eight and eleven times. Inlet water chemistry samples were considered as representative of upstream catchment water chemistry while outlet samples were representative of water chemistry as modified by wetland biogeochemical processes and delivered to downstream receiving waters. Samples for dissolved GHG analysis ( $\text{CH}_4$  and  $\text{N}_2\text{O}$ ) were collected during each sampling event from the center of each wetland. Additionally, we measured total  $\text{CH}_4$  emissions (i.e. diffusive + ebullitive) from a subset of wetlands on five occasions: twice during fall/winter and three times during summer. Sediment cores were collected and processed in summers 2020–2021 from inlets and outlets of all wetlands.

### 3. Analytical methods

DO, EC, atmospheric pressure, water temperature, chlorophyll-a and pH were measured *in-situ* using a Hanna Instruments Multiparameter Meter HI 9829 and Turner Designs FluoroSense. All other water chemistry parameters were analyzed at the Department of Aquatic Sciences and Assessment, Swedish University of Agricultural Sciences (SLU) accredited laboratories (Fölster *et al* 2014, Wallman and Demandt 2026). Sediment cores were processed following protocols described by Lannergård *et al* (2020). Cores were sectioned in the field and the top 4 cm retained for further analysis. Loss on ignition was used as a proxy for sediment organic matter content (Henry *et al* 2024). Sediment P fractionation used a sequential method following Psenner and Pucsko (1988), adapted by Hupfer *et al* 2009). Briefly, the method characterized five P pools having different degrees of sediment association, ranging from loosely to tightly bound fractions (see Lannergård *et al* 2020 for further information). We define the non-labile P fraction (NLP\_Sed) as the sum of organically bound, aluminum bound and calcium bound P. TotP\_Sed was the sum of all sediment P fractions. Sediment P concentrations are reported on a volumetric basis ( $\text{mg cm}^{-3}$ ), i.e. mass of P per unit volume of fresh sediment. We report volumetric units instead of the more commonly reported dry weights as the former offer a more accurate representation of the size of

the sediment P store. Here, we report average values derived from all sediment cores collected at each wetland.

Dissolved inorganic carbon (DIC) and excess partial pressure of  $\text{CO}_2$  ( $\text{epCO}_2$ ) were estimated from field measurements of water temperature, atmospheric pressure and pH, and laboratory alkalinity measurements using routines from Jarvie *et al* (2017).

Dissolved  $\text{CH}_4$  and  $\text{N}_2\text{O}$  concentrations were analyzed by the headspace method (Kling *et al* 1992). Briefly, 30 ml of water was equilibrated with 30 ml of ambient atmospheric air in a 60 ml syringe by shaking for one minute in the field. Headspace corresponding to 20 ml was then injected into a 20 ml glass vial through a butyl-PTFE septum in an Al cap.  $\text{CH}_4$  and  $\text{N}_2\text{O}$  concentrations were analyzed on a Clarus 500 Perkin Elmer gas chromatograph, fitted with flame ionization and electron capture detectors. Dissolved concentrations were calculated using Henry's Law and solubility functions for  $\text{CH}_4$  and  $\text{N}_2\text{O}$  (Wiesenburg and Guinasso 1979, Weiss and Price 1980).  $\text{CH}_4$  emissions were measured using multiple (3–9) floating chambers deployed for 24h. Simultaneously, we measured dissolved  $\text{CH}_4$  concentrations. Methods of Bastviken *et al* (2004) were used to estimate ebullitive  $\text{CH}_4$  flux.

### 4. Statistical analysis

Water chemistry measurements were summarized by season (summer *vs* fall/winter), location (inlet or outlet), and season by location (table 1). Geometric means were used in all cases to summarize results. Differences in mean values were assessed using paired *t*-tests. Spearman correlations were calculated for water chemistry, GHG and sediment parameters. An absolute Spearman value  $>0.5$  corresponding to a nominal *p* value of  $\sim 0.001$  at 39 df was used as a threshold for reporting meaningful correlations. All other statistical significance estimates report nominal *p*-values (i.e. not adjusted for multiple comparisons). Principal components analyses (PCA) were performed to visualize patterns in the data. All analyses were performed in R 4.5.1 using code from the `vegan()` library.

### 5. Results

Overall, seasonal differences in water chemistry and dissolved GHG concentrations were greater than differences between inlets and outlets, with average absolute *t*-values of 2.8 and 2.4 for inlets and outlets respectively compared to average *t*-values of 1.0 for inlets versus outlets in both seasons (table 1).

**Table 1.** Results of paired *t*-test analyses evaluating seasonal and inlet/outlet differences in water chemistry and modeled GHG concentrations. Nominally significant differences are shown in bold. The following parameters are reported: alkalinity (Alk, meq l<sup>-1</sup>), ammonium (NH<sub>4</sub>, mg N l<sup>-1</sup>), chlorophyll-a (Chl, ug l<sup>-1</sup>), modeled CO<sub>2</sub> saturation (epCO<sub>2</sub>, dimensionless), conductivity (Cond, ms m<sup>-1</sup> @ 25 C), modeled dissolved inorganic carbon concentration (DIC, mg l<sup>-1</sup>), dissolved oxygen concentration (DO, mg l<sup>-1</sup>), phosphate (PO<sub>4</sub>, mg P l<sup>-1</sup>), filtered total phosphorus (FiltP, mg P l<sup>-1</sup>), nitrate (NO<sub>3</sub>, mg N l<sup>-1</sup>), pH (pH, dimensionless), suspended sediments (SS, mg l<sup>-1</sup>), total nitrogen (TotN, mg N l<sup>-1</sup>), total organic carbon (TOC, mg l<sup>-1</sup>), total phosphorus (TotP, mg P l<sup>-1</sup>).

| Fall/Winter Inlet vs. Outlet                |                            |              |                  |               |
|---------------------------------------------|----------------------------|--------------|------------------|---------------|
| Parameter                                   | Units                      | Inlet Mean   | Outlet Mean      | t_statistic   |
| Alkalinity                                  | meq l <sup>-1</sup>        | 1.58         | 1.57             | 0.28          |
| Ammonium as N                               | mg N l <sup>-1</sup>       | 0.06         | 0.07             | -1.02         |
| <b>Chlorophyll-a</b>                        | <b>ug l<sup>-1</sup></b>   | <b>7.45</b>  | <b>9.44</b>      | <b>-2.19</b>  |
| CO <sub>2</sub> supersaturation ratio       | dimensionless              | 14.57        | 10.1             | 1.44          |
| Conductivity                                | ms/m @ 25 C                | 38.24        | 36.11            | 1.41          |
| Dissolved inorganic carbon                  | mg l <sup>-1</sup>         | 23.27        | 21.56            | 1.30          |
| Dissolved oxygen                            | mg l <sup>-1</sup>         | 8.5          | 7.62             | 2.00          |
| Filtered phosphate as P                     | mg P l <sup>-1</sup>       | 0.04         | 0.04             | 0.04          |
| Filtered total phosphorus                   | mg P l <sup>-1</sup>       | 0.05         | 0.05             | -0.33         |
| Nitrate as N                                | mg N l <sup>-1</sup>       | 4.27         | 3.87             | 1.84          |
| pH                                          | dimensionless              | 7.28         | 7.32             | -0.59         |
| Suspended sediment                          | mg l <sup>-1</sup>         | 44.21        | 56.62            | -1.15         |
| Total nitrogen                              | mg N l <sup>-1</sup>       | 5.34         | 5.03             | 1.42          |
| Total organic carbon                        | mg l <sup>-1</sup>         | 10.68        | 12.92            | -1.50         |
| Total Phosphorus                            | mg P l <sup>-1</sup>       | 0.15         | 0.2              | -1.70         |
| Summer Inlet vs Outlet                      |                            |              |                  |               |
| Parameter                                   | Units                      | Inlet Mean   | Outlet Mean      | t_statistic   |
| Alkalinity                                  | meq l <sup>-1</sup>        | 2.06         | 2.11             | -0.61         |
| Ammonium as N                               | mg N l <sup>-1</sup>       | 0.17         | 0.08             | 1.02          |
| <b>Chlorophyll-a</b>                        | <b>ug l<sup>-1</sup></b>   | <b>9.33</b>  | <b>14.08</b>     | <b>-2.57</b>  |
| CO <sub>2</sub> supersaturation ratio       | dimensionless              | 5.52         | 5.73             | -0.19         |
| <b>Conductivity</b>                         | <b>ms/m @ 25 C</b>         | <b>44.3</b>  | <b>40.72</b>     | <b>2.37</b>   |
| Dissolved inorganic carbon                  | mg l <sup>-1</sup>         | 25.81        | 26.36            | -0.52         |
| <b>Dissolved oxygen</b>                     | <b>mg l<sup>-1</sup></b>   | <b>1.86</b>  | <b>1.34</b>      | <b>4.21</b>   |
| Filtered phosphate as P                     | mg P l <sup>-1</sup>       | 0.06         | 0.05             | 0.71          |
| Filtered total phosphorus                   | mg P l <sup>-1</sup>       | 0.08         | 0.07             | 0.46          |
| <b>Nitrate as N</b>                         | <b>mg N l<sup>-1</sup></b> | <b>2.39</b>  | <b>1.77</b>      | <b>2.10</b>   |
| pH                                          | dimensionless              | 7.78         | 7.83             | -0.65         |
| Suspended sediment                          | mg l <sup>-1</sup>         | 21.84        | 22               | -0.07         |
| <b>Total nitrogen</b>                       | <b>mg N l<sup>-1</sup></b> | <b>4.06</b>  | <b>3.21</b>      | <b>2.55</b>   |
| <b>Total organic carbon</b>                 | <b>mg l<sup>-1</sup></b>   | <b>11.56</b> | <b>13</b>        | <b>-2.58</b>  |
| Total Phosphorus                            | mg P l <sup>-1</sup>       | 0.2          | 0.2              | -0.38         |
| Inlet Summer vs. Fall/Winter                |                            |              |                  |               |
| Parameter                                   | Units                      | Summer Mean  | Fall/Winter Mean | t_statistic   |
| <b>Alkalinity</b>                           | <b>meq l<sup>-1</sup></b>  | <b>2.06</b>  | <b>1.61</b>      | <b>3.66</b>   |
| Ammonium as N                               | mg N l <sup>-1</sup>       | 0.17         | 0.06             | 1.20          |
| <b>Chlorophyll-a</b>                        | <b>ug l<sup>-1</sup></b>   | <b>9.33</b>  | <b>7.45</b>      | <b>2.03</b>   |
| <b>CO<sub>2</sub> supersaturation ratio</b> | <b>dimensionless</b>       | <b>5.52</b>  | <b>14.82</b>     | <b>-2.80</b>  |
| Conductivity                                | ms/m @ 25 C                | 44.3         | 38.8             | 1.59          |
| Dissolved inorganic carbon                  | mg l <sup>-1</sup>         | 25.81        | 23.66            | 1.12          |
| <b>Dissolved oxygen</b>                     | <b>mg l<sup>-1</sup></b>   | <b>1.86</b>  | <b>8.5</b>       | <b>-17.13</b> |
| Filtered phosphate as P                     | mg P l <sup>-1</sup>       | 0.06         | 0.04             | 1.66          |
| <b>Filtered total phosphorus</b>            | <b>mg P l<sup>-1</sup></b> | <b>0.08</b>  | <b>0.05</b>      | <b>2.27</b>   |
| Nitrate as N                                | mg N l <sup>-1</sup>       | 2.39         | 4.34             | -3.89         |
| <b>pH</b>                                   | <b>dimensionless</b>       | <b>7.78</b>  | <b>7.28</b>      | <b>7.85</b>   |
| <b>Suspended sediment</b>                   | <b>mg l<sup>-1</sup></b>   | <b>21.84</b> | <b>35.6</b>      | <b>-2.26</b>  |
| <b>Total nitrogen</b>                       | <b>mg N l<sup>-1</sup></b> | <b>4.06</b>  | <b>5.41</b>      | <b>-3.04</b>  |
| Total organic carbon                        | mg l <sup>-1</sup>         | 11.56        | 10.57            | 1.35          |
| Total phosphorus                            | mg P l <sup>-1</sup>       | 0.2          | 0.14             | 1.75          |

(Continued.)

Table 1. (Continued.)

| Outlet Summer vs Fall/Winter                |                            |              |                  |               |
|---------------------------------------------|----------------------------|--------------|------------------|---------------|
| Parameter                                   | Units                      | Summer Mean  | Fall/Winter Mean | t_statistic   |
| <b>Alkalinity</b>                           | <b>meq l<sup>-1</sup></b>  | <b>2.09</b>  | <b>1.57</b>      | <b>3.89</b>   |
| Ammonium as N                               | mg N l <sup>-1</sup>       | 0.08         | 0.07             | 0.73          |
| <b>Chlorophyll-a</b>                        | <b>ug l<sup>-1</sup></b>   | <b>14.08</b> | <b>9.44</b>      | <b>3.00</b>   |
| <b>CO<sub>2</sub> supersaturation ratio</b> | <b>dimensionless</b>       | <b>5.76</b>  | <b>10.1</b>      | <b>-3.99</b>  |
| Conductivity                                | ms/m @ 25 C                | 40.22        | 36.11            | 1.41          |
| <b>Dissolved inorganic carbon</b>           | <b>mg l<sup>-1</sup></b>   | <b>26.12</b> | <b>21.56</b>     | <b>2.86</b>   |
| <b>Dissolved oxygen</b>                     | <b>mg l<sup>-1</sup></b>   | <b>1.34</b>  | <b>7.62</b>      | <b>-16.42</b> |
| Filtered phosphate as P                     | mg P l <sup>-1</sup>       | 0.05         | 0.04             | 0.61          |
| Filtered total phosphorus                   | mg P l <sup>-1</sup>       | 0.07         | 0.05             | 1.67          |
| <b>Nitrate as N</b>                         | <b>mg N l<sup>-1</sup></b> | <b>1.73</b>  | <b>3.87</b>      | <b>-5.08</b>  |
| <b>pH</b>                                   | <b>dimensionless</b>       | <b>7.83</b>  | <b>7.32</b>      | <b>9.36</b>   |
| <b>Suspended sediment</b>                   | <b>mg l<sup>-1</sup></b>   | <b>23.28</b> | <b>56.62</b>     | <b>-2.85</b>  |
| <b>Total nitrogen</b>                       | <b>mg N l<sup>-1</sup></b> | <b>3.16</b>  | <b>5.03</b>      | <b>-4.93</b>  |
| Total organic carbon                        | mg l <sup>-1</sup>         | 12.98        | 12.92            | 0.04          |
| Total phosphorus                            | mg P l <sup>-1</sup>       | 0.22         | 0.2              | 0.84          |

Note: Bold text indicates a significant difference at a nominal p value of 0.05.

The largest seasonal differences in water chemistry were for inlet DO concentrations, which averaged 1.86 mg l<sup>-1</sup> in summer and 8.5 mg l<sup>-1</sup> in fall/winter (table 1). Outlet DO concentrations followed a similar pattern with average summer concentrations of 1.34 mg l<sup>-1</sup> versus 7.62 mg l<sup>-1</sup> in fall/winter. Summer pH was notably higher than fall/winter in both inlets (7.78 vs 7.28) and outlets (7.83 vs 7.32). Outlet nitrate and total N concentrations were lower in summer (1.73 and 3.16 mg l<sup>-1</sup>) versus fall/winter (3.87 and 5.03 mg l<sup>-1</sup>). Summer inlet nitrate and total N concentrations (2.39 mg l<sup>-1</sup> and 4.06 mg l<sup>-1</sup> respectively) were lower than fall/winter values (4.34 mg l<sup>-1</sup> and 5.41 mg l<sup>-1</sup>). Summer outlet chlorophyll-a was higher than corresponding fall/winter values (14.08 vs 9.45 µg l<sup>-1</sup>). Summer alkalinity was higher for both inlets and outlets (averages of 2.1 mg l<sup>-1</sup> vs 1.6 mg l<sup>-1</sup>).

The largest difference between inlet and outlet water chemistry was for summer DO concentrations, with average inlet and outlet values of 1.9 mg l<sup>-1</sup> and 1.3 mg l<sup>-1</sup> respectively. Other water chemistry parameters with differences between summer inlet and values were TOC (11.6 mg l<sup>-1</sup> vs 13.0 mg l<sup>-1</sup>), chlorophyll (9.1 vs 14.1 µg l<sup>-1</sup>) and total N (4.1 vs 3.2 mg l<sup>-1</sup>). There were no differences between inlet and outlet concentrations for any P parameters in either season.

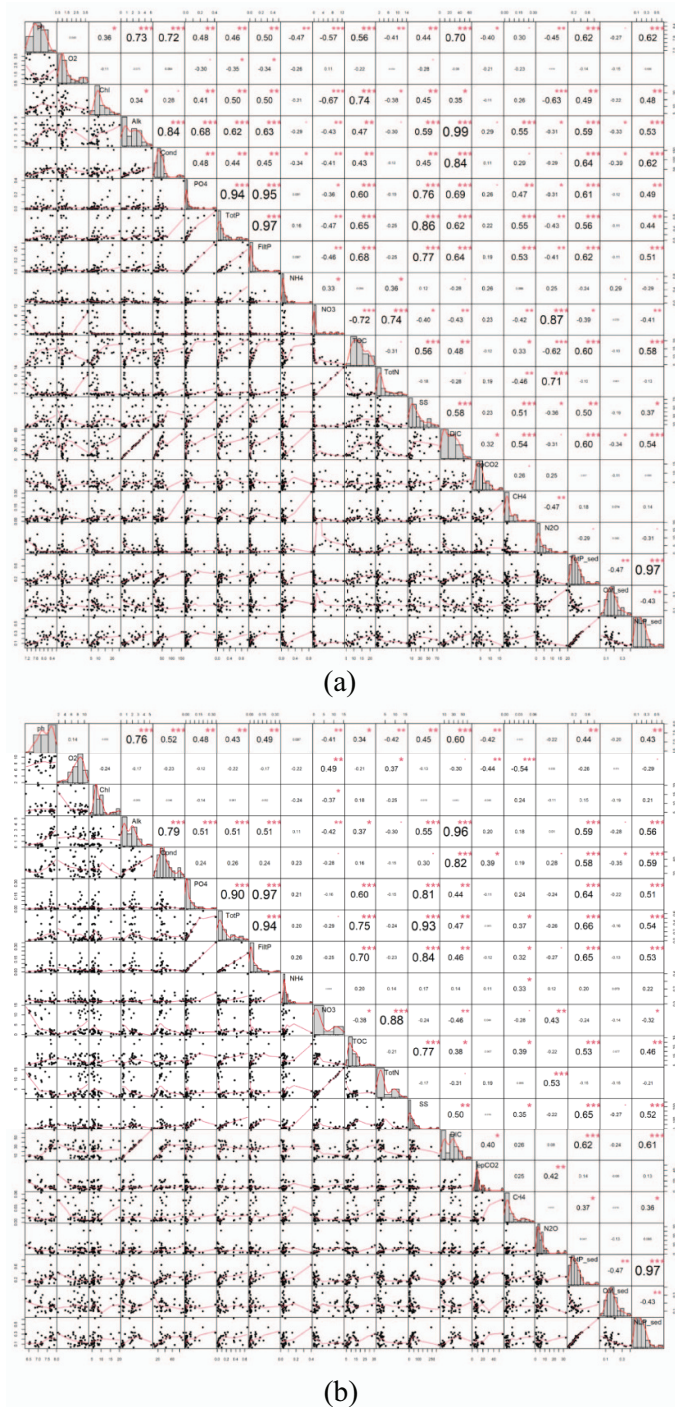
Summer dissolved GHG concentrations were, in some cases, related to water chemistry. Dissolved CH<sub>4</sub> concentrations were positively correlated to total and filtered P, suspended sediments, DIC and alkalinity (figure 2(a)). N<sub>2</sub>O concentrations were positively correlated to nitrate and total N and negatively to chlorophyll-a and TOC. CO<sub>2</sub> (as epCO<sub>2</sub>) was not strongly correlated with any other parameters (figure 2(a)).

A different pattern occurred during fall/winter (figure 2(b)). Dissolved CH<sub>4</sub> was negatively correlated with DO and unrelated to either total N or N<sub>2</sub>O. N<sub>2</sub>O concentrations were correlated to total N and had weak positive relationships with nitrate and epCO<sub>2</sub>. Neither summer nor fall/winter dissolved GHG concentrations were related to sediment properties (figures 2(a) and (b)).

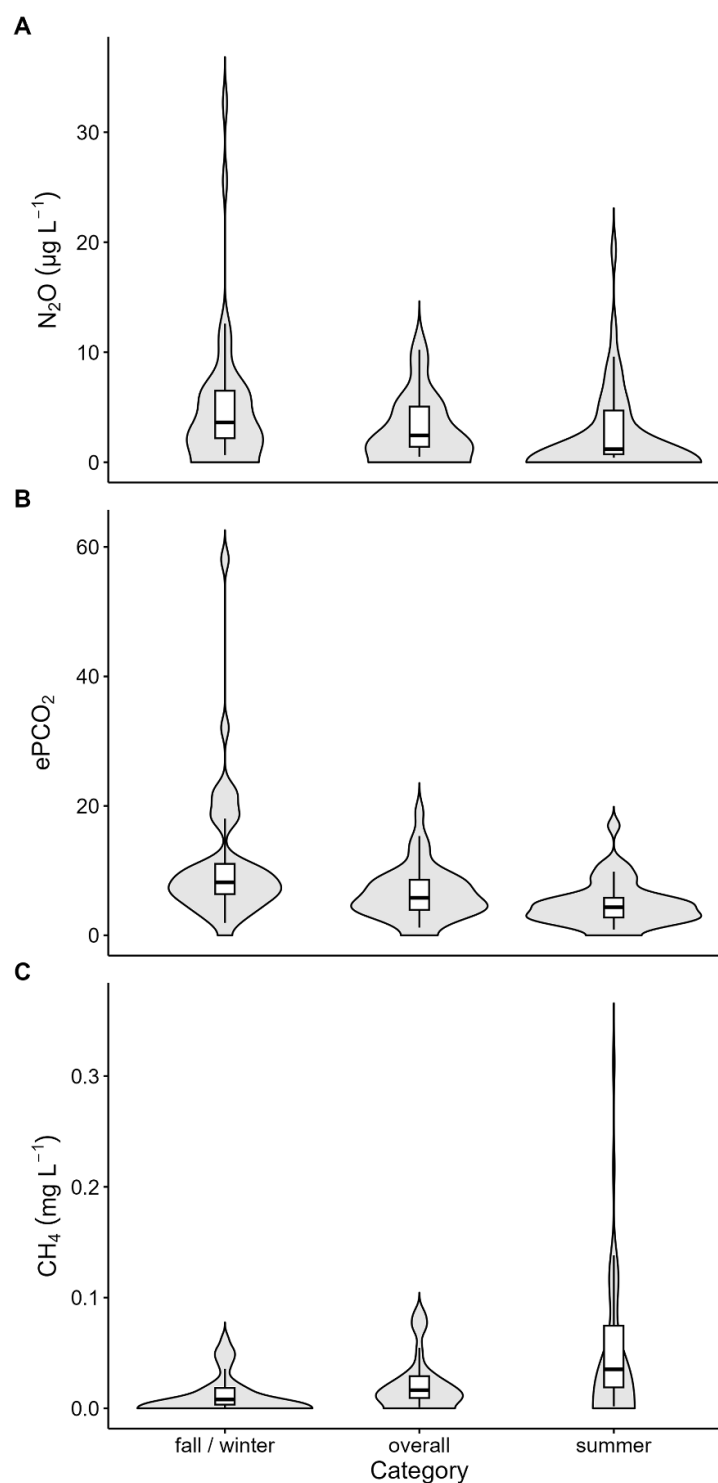
All three P parameters correlated strongly with each other, TOC, suspended sediments and DIC in both seasons. Total and filtered P were more strongly correlated than phosphate P with chlorophyll-a in summer, but there were no relationships between any P parameters and chlorophyll-a during fall/winter. Correlations were stronger between water column P and dissolved CH<sub>4</sub> in summer than fall/winter. The negative correlations between phosphorus, nitrate and total N were stronger during summer than fall/winter (figures 2(a) and (b)).

There were large seasonal differences in total CH<sub>4</sub> emissions (16.4 vs 9.2 mg m<sup>-2</sup> d<sup>-1</sup> for summer and fall/winter, nominal  $p = 0.042$ ). The percentage contribution of ebullition to total flux also differed between seasons (47% for summer, 75% for fall/winter, nominal  $p < 0.01$ ) (table 2). Average summer dissolved CH<sub>4</sub> concentrations were higher than fall/winter values (30.1 vs 8.8 µg l<sup>-1</sup>). An opposite pattern was seen for N<sub>2</sub>O where summer concentrations were lower than those observed in fall/winter (1.8 vs 3.7 µg l<sup>-1</sup>). Average summer epCO<sub>2</sub> values were 4.1, compared to 8.5 during fall/winter (figure 3, table 2).

Ammonium concentrations did not correlate strongly with any parameter during either season. Nitrate and total N were positively correlated in both seasons. Summer nitrate concentrations were negatively correlated with phosphorus, chlorophyll-a,



**Figure 2.** (a): Spearman correlation plots for summer water chemistry, dissolved greenhouse gas (GHG) and sediment measurements. Water chemistry measurements are pooled across inlets and outlets. In all cases, geometric mean values are reported. From top to bottom, the parameters are pH,  $O_2$  (dissolved oxygen concentration), Chl (chlorophyll-a concentration), Alk (alkalinity), Cond (conductivity),  $PO_4$  (phosphate concentration as P), TotP (total phosphorus concentration as P), FiltP (filtered total phosphorus concentration as P),  $NH_4$  (ammonia concentration as N),  $NO_3$  (nitrate concentration as N), TOC (total organic carbon concentration), TotN (total nitrogen concentration as N), SS (suspended solids concentration), DIC (modeled dissolved inorganic carbon concentration),  $epCO_2$  (modeled equilibrium partial pressure of  $CO_2$ ),  $CH_4$  (dissolved methane concentration),  $N_2O$  (dissolved nitrous oxide concentration), TotP\_Sed (sediment total phosphorus concentration as P), OM\_Sed (sediment organic matter concentration as estimated from loss on ignition) and NLP\_Sed (concentration of non-labile phosphorus fractions in the sediment as P). (b): Spearman correlation plots for fall/winter water chemistry, dissolved greenhouse gas (GHG) and sediment measurements. Water chemistry measurements are pooled across inlets and outlets. In all cases, geometric mean values are reported. From top to bottom, the parameters are pH,  $O_2$  (dissolved oxygen concentration), Chl (chlorophyll-a concentration), Alk (alkalinity), Cond (conductivity),  $PO_4$  (phosphate concentration as P), TotP (total phosphorus concentration as P), FiltP (filtered total phosphorus concentration as P),  $NH_4$  (ammonia concentration as N),  $NO_3$  (nitrate concentration as N), TOC (total organic carbon concentration), TotN (total nitrogen concentration as N), SS (suspended solids concentration), DIC (modeled dissolved inorganic carbon concentration),  $epCO_2$  (modeled equilibrium partial pressure of  $CO_2$ ),  $CH_4$  (dissolved methane concentration),  $N_2O$  (dissolved nitrous oxide concentration), TotP\_Sed (sediment total phosphorus concentration as P), OM\_Sed (sediment organic matter concentration as estimated from loss on ignition) and NLP\_Sed (concentration of non-labile phosphorus fractions in the sediment as P).



**Figure 3.** Violin plot showing summer, fall /winter and overall concentrations of dissolved  $\text{N}_2\text{O}$ , dissolved  $\text{CH}_4$  and equilibrium partial pressure of  $\text{CO}_2$  ( $\text{ePCO}_2$ ).

and TOC but positively correlated to  $\text{N}_2\text{O}$ . Winter nitrate concentrations were positively correlated with  $\text{O}_2$  and negatively with DIC. Total N concentrations were negatively correlated with dissolved  $\text{CH}_4$  in both seasons.

Summer chlorophyll concentrations were positively correlated with all P parameters and negatively with nitrate and total N. Winter chlorophyll-a was negatively correlated with nitrate. Geometric mean

summer inlet molar N:P ratios were 11 and 81 for the eastern and western sites respectively. These values are either side of the 16:1 N:P Redfield ratio, suggesting that primary productivity in eastern wetlands is more likely to be N limited while western locations are more likely P limited.

Sediment OC levels were only related to sediment total P values. Sediment total P was positively correlated to water column P, TOC, suspended matter

**Table 2.** Results of paired *t*-test analyses evaluating seasonal differences in dissolved GHG concentrations. All tests had 39 degrees of freedom. Nominal *p*-values are reported.

| Variable                                                                         | Summer                 |                        | Fall/winter            |                        | t_statistic | Significant |
|----------------------------------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|-------------|-------------|
|                                                                                  | Mean                   | SD                     | Mean                   | SD                     |             |             |
| epCO <sub>2</sub>                                                                | $4.98 \times 10^{00}$  | $3.27 \times 10^{00}$  | $1.12 \times 10^{01}$  | $1.00 \times 10^{01}$  | −3.97       | ***         |
| Dissolved CH <sub>4</sub><br>(mg l <sup>−1</sup> )                               | $5.62 \times 10^{-02}$ | $6.33 \times 10^{-02}$ | $1.54 \times 10^{-02}$ | $1.70 \times 10^{-02}$ | 4.23        | ***         |
| Dissolved N <sub>2</sub> O<br>(ug l <sup>−1</sup> )                              | $3.19 \times 10^{00}$  | $3.94 \times 10^{00}$  | $5.66 \times 10^{00}$  | $6.29 \times 10^{00}$  | −2.17       | *           |
| Total CH <sub>4</sub> Emissions<br>(mg m <sup>−2</sup> d <sup>−1</sup> )         | $3.75 \times 10^{01}$  | $3.18 \times 10^{01}$  | $2.69 \times 10^{01}$  | $3.33 \times 10^{01}$  | −2.19       | *           |
| Ebullitive CH <sub>4</sub><br>emissions<br>(mg m <sup>−2</sup> d <sup>−1</sup> ) | $2.10 \times 10^{01}$  | $2.27 \times 10^{01}$  | $2.16 \times 10^{01}$  | $2.77 \times 10^{01}$  | 0.15        |             |

and DIC in both seasons. Sediment non-labile P was strongly correlated with sediment total P and was also positively related to water column P and TOC.

PCAs performed on summer and fall/winter water chemistry data corroborated the patterns reported above (supplementary figures 1(a) and (b)). In both seasons, the first two PCA axes explained approximately half of the total variation. In the summer biplot (supplementary figure 1(a)), total N, nitrate and N<sub>2</sub>O were closely related to each other, and dissolved CH<sub>4</sub> concentrations were closely related to water column P. Sediment P concentrations were independent of water column P but correlated to DIC, alkalinity and conductivity (supplementary figure 1(a)). In both seasons, the three GHGs were largely independent of each other, as were sediment P fractions and organic matter. In winter, water column P was correlated with ammonium. Winter N<sub>2</sub>O was unrelated to nitrate or total N, winter CH<sub>4</sub> was unrelated to water column P (supplementary figure 1(b)).

## 6. Discussion

In our 40 wetlands across two regions, seasonal differences in water chemistry were typically greater than within-season differences between inlets and outlets. This suggests that upstream catchment drivers (e.g. land use, hydrology) have a larger influence on downstream water chemistry than in-wetland processes. Higher winter nitrate concentrations are likely indicative of lower plant uptake during the dormant season (Bastviken *et al* 2009). Fall/winter dissolved GHG concentrations were not related to nutrient concentrations, which may reflect lower levels of microbial activity during cooler times of the year (Ollivier *et al* 2019, Liu *et al* 2025). However, ebullitive CH<sub>4</sub> fluxes were not influenced by season.

Unsurprisingly, chlorophyll-a levels in the wetlands were higher in summer than fall/winter. Differences between inlet and outlet DOC concentrations were also higher in summer, which could

indicate higher rates of autochthonous primary production resulting in a net export of TOC. Notably, residence time is a key control on net OC source/sink behavior in such wetlands (Jones *et al* 2025). The difference in summer inlet and outlet DO is also suggestive of higher levels of microbial activity during summer. This was related to higher dissolved CH<sub>4</sub> concentrations, and higher total (diffusive + ebullitive) CH<sub>4</sub> emissions, during summer as is frequently seen in agricultural wetlands and ponds (e.g. Stadmark and Leonardson 2005, Mander *et al* 2024). Despite the higher summer CH<sub>4</sub> emissions, fall/winter emissions were non-negligible, and the percentage contribution of ebullition to total flux was greater during the cold season, highlighting the need for year-round monitoring (e.g. Liu *et al* 2025). In contrast to CH<sub>4</sub>, dissolved N<sub>2</sub>O concentrations and epCO<sub>2</sub> levels were higher in winter than in summer. Higher N<sub>2</sub>O concentrations may be associated with elevated dissolved N inputs during winter associated with less plant uptake in the catchment. Solubility of CO<sub>2</sub> is negatively related to temperature, and higher primary productivity (as suggested by the higher summer chlorophyll-a) will consume more CO<sub>2</sub> (Stadmark and Leonardson 2005).

Phosphorus concentrations were negatively related to P<sub>AL</sub>, a measure of plant available P. This is a somewhat surprising finding as it might be expected that higher aquatic P concentrations should be associated with more plant available P in the soil. However, instream P concentrations may be more closely related to riparian and stream sediment P concentrations, which can be higher than concentrations in surrounding agricultural fields (Hoffman *et al* 2009). Concentrations are also likely driven by hydrological events. P<sub>AL</sub> was higher in western catchments whereas water P concentrations were higher in eastern wetlands. This region is characterized by erodible fine-textured soils, despite lower soil P<sub>AL</sub> content. Even soils with relatively low legacy soil P content can produce significant P loads due to the influence of hydrology, overland flow and soil erosion (Kleinman *et al* 2011, Djodjic *et al* 2018).

Higher nutrient inputs were correlated with higher absolute levels of nutrient removal and retention. Differences between inlet and outlet N concentrations were positively related to influent N concentrations. Although there was no apparent difference in inlet and outlet concentrations for P, water column and sediment P were well correlated, suggesting a connection between loading and retention. From a monitoring perspective, the high correlation between recalcitrant (NLP\_Sed) and total sediment phosphorus (TotP\_Sed) was encouraging as it suggests P retention can be estimated using simpler methods without needing fractionation. Future wetland monitoring should also evaluate sediment N storage.

The weak negative correlation between sediment P and OC may indicate tradeoffs between P removal and C accumulation. This contrasts with patterns typically reported from natural ecosystems (Reddy *et al* 1999) and suggests decoupling of sediment P and C burial. At least for these constructed wetlands, higher levels of P removal have a double climate impact: input P concentrations and dissolved CH<sub>4</sub> levels correlate positively and higher sediment P concentrations are associated with lower sediment OC concentrations. This apparent tradeoff should be explored further. As we lacked information on age and maintenance (e.g. dredging or sediment removal) history of all wetlands, we could not estimate sediment P and organic matter accumulation rates. Concentrations of OC and P in the uppermost depths of the sediment profile could be a proxy for sequestration rates if higher concentrations are indicative of higher accumulation rates.

Seasonality should be considered when assessing tradeoffs. P inputs to wetlands occur mainly during the wet season (fall/winter) when correlations between P and CH<sub>4</sub> are weak at best (figure 2(b)). Additionally, CH<sub>4</sub> and O<sub>2</sub> are negatively correlated in fall/winter. In summer (figure 2(a)), when flows are low, but P concentrations are high. CH<sub>4</sub> and P are correlated but the correlation to O<sub>2</sub> is weak. As summer O<sub>2</sub> concentrations are generally low (1.86 and 1.34 mg l<sup>-1</sup> for inlets and outlets respectively), aeration could potentially reduce CH<sub>4</sub> emissions.

Our findings are likely to be transferable to constructed wetlands in agricultural regions with pronounced seasonality and moderate nutrient loadings, e.g. Europe (Rizzo *et al* 2023) and Canada (Jensen *et al* 2022). Warm, dry regions (e.g. Australia, Webb *et al* 2023), ponds with extremely high nutrient loadings, e.g. for aquaculture (e.g. Yang *et al* 2024) or saline systems (e.g. Soued *et al* 2024) may behave differently to the systems presented here.

Downstream ES delivery is influenced by multiple factors including upstream nutrient loading and in-wetland processing. Measures to reduce upstream

nutrient inputs including, e.g. buffer zones, structural liming, spring tillage, catch crops, etc (Sandström *et al* 2024) can also reduce wetland ability to mitigate eutrophication but could also lower GHG emissions. Furthermore, maintenance improves wetland ES delivery (Geranmayeh *et al* 2025). Vegetation harvesting and dredging both contribute to C removal. Dredging contributes directly to P removal and can increase residence times, thereby promoting effective nutrient retention.

NBS use nature and natural processes, but their effectiveness depends on appropriate design, siting and maintenance. Higher nutrient loads are associated with more effective retention and/or removal but are also more likely to result in pollution swapping. Nitrogen inputs are positively correlated with dissolved N<sub>2</sub>O concentrations and higher water column P values may be indicative of more effective sediment P retention but are also positively related to dissolved CH<sub>4</sub>. The existence of such tradeoffs should not preclude the use of constructed wetlands as nature positive solutions to the pressing societal problems caused by surface water eutrophication.

## 7. Conclusions

The large seasonal differences in inlet water chemistry suggest a strong influence of climate and land management on nutrient fluxes. There was some evidence of nitrogen concentration reductions, but overall, differences between inlet and outlet water chemistry were smaller than seasonal differences in concentrations. The positive relationship between influent P concentrations and sediment P suggests concentration-dependent retention. Positive relationships between influent nutrient concentrations and GHG concentrations suggest tradeoffs between mitigating eutrophication and climate impacts.

## Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.6084/m9.figshare.31899763>.

Supplementary data 1 available at <https://doi.org/10.1088/1748-9326/ae9492/data1>.

Supplementary data 2 available at <https://doi.org/10.1088/1748-9326/ae9492/data2>.

Supplementary data 3 available at <https://doi.org/10.1088/1748-9326/ae9492/data3>.

Supplementary data 4 available at <https://doi.org/10.1088/1748-9326/ae9492/data4>.

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